

A General Linear-Regression Analysis Applied to the 3-Parameter Weibull Distribution

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Key Words — Nonlinear regression, Weibull distribution

Reader Aids —

General purpose: Present Weibull and Weibull-like models
Special math needed for explanations: Elementary probability
Special math needed to use results: Same
Results useful to: Reliability analysts & theoreticians

Summary & Conclusions — The conventional techniques of linear regression analysis (linear least squares) applied to the 3-parameter Weibull distribution are extended (not modified), and new techniques are developed for the 3-parameter Weibull distribution. The three pragmatic estimation methods in this paper are simple, accurate, flexible, and powerful in dealing with difficult problems such as estimates of the 3 parameters becoming non-positive. In addition, the inherent disadvantages of the 3-parameter Weibull distribution are revealed; the advantages of a new 3-parameter Weibull-like distribution over the original Weibull distribution are explored; and the potential of a 4-parameter Weibull-like distribution is briefly mentioned.

This paper demonstrates how a general linear regression-analysis or linear least-squares breaks away from the classical or modern nonlinear regression-analysis or nonlinear least-squares. By adding a parameter to the simplest 2-parameter linear regression model (AB-model), two kinds of ABC-models (elementary 3-parameter nonlinear regression models) are found, and then a 4-parameter AABC-model is built as an example of multi-parameter nonlinear regression models. Although some other techniques are still necessary, additional applications of the ABC-models are strongly implied.

1. INTRODUCTION

Acronyms¹

AB-model	model with a, b as parameters, see (1-5)
ABC-model	model with a, b, c as parameters, see (1-6) - (1-7)
AABC-model	model with a_1, a_2, b, c as parameters, see (1-8)
LRA, LLS	linear [regression analysis, least squares]
NLRA, NLLS	nonlinear [regression analysis, least squares]
GLRA, GLLS	general linear [regression analysis, least squares]
2PW, 3PW _{org}	[2, 3]-parameter Weibull distribution
3PW _{new} , 4PW	[3, 4]-parameter Weibull-like ² distribution.

¹The singular & plural of an acronym are always spelled the same.

The 2PW Cdf can be transformed into a straight line by a double logarithmic transformation. This allows for a direct application of a LRA or LLS. However, the 3PW_{org} Cdf can not be transformed into a straight line by conventional techniques. It seems that no method in the literature tries to apply either classical or modern NLRA or NLLS to the 3PW_{org} Cdf, other than the graphical method where the location parameter must be approximated several times. This latter method appears in most books for reliability engineers. Hence, the methods in this paper fill the gap by, for example:

- improving the graphical method,
- providing simple, approximate and non-approximate, estimation methods,
- analyzing the problems of 3PW_{org} that are ignored by others.

There are many advantages of LRA/LLS & NLRA/NLLS. But the unique one is:

- a reliability engineer, especially when handling the field data instead of the expensive laboratory data, can simply relocate³ those outliers (if there are enough data to do so) without worrying about anything. This advantage is facilitated by the new methods in this paper.

By observing how an additional parameter makes the simplest 2-parameter linear regression model into a nonlinear model, a new 3PW_{new} is presented. Compared with this new one, the original 3PW_{org} is neither super-accurate nor free from inherent disadvantages; though the new one will never be perfect itself. To improve this situation, the 3PW_{org} & 3PW_{new} can either be used simultaneously to cover the other's deficiencies or can be combined into 4PW. Numerous papers propose estimation methods and compete with each other for being "even more" accurate.

To search for new pragmatic methods & techniques, this paper proposes a GLRA & GLLS. The ideas of GLRA & GLLS are developed in a forthright way:

- Add another parameter (to a multi-parameter nonlinear regression model) to open a new area (in applying the simple 2-parameter linear regression model).

²Statisticians generally reserve the name, *Weibull*, for a distribution that can be transformed by a simple location or scale parameter change into the original Weibull equation that belongs to the extreme-value family of distributions. Thus for my extensions that do not meet that strict criterion, the name, *Weibull-like* is used to avoid possible confusion.

³In a competent statistical analysis, data are never actually discarded or rejected. Rather, they are relocated into another data-pool for further engineering & statistical examination as to their meaning for the real problem at hand.

A unique advantage is:

- a reliability engineer does not need to be involved in all of those unfamiliar and/or bothersome nonlinear system problems which can lead to strange solutions.

Though further research has been pursued and many valuable achievements have been made, GLRA & GLLS are still young buds that can grow into a new research branch under the care of reliability analysts & theoreticians.

As is true for other graphical methods, the ones in this paper do not provide any measure of uncertainty in the parameter estimates. Nor are any measures of uncertainty shown for the analytic estimates; such measures could be developed.

The 2PW & 3PW_{org} Cdf's are:

$$F(t; \alpha, \beta) = 1 - \exp(-t^\beta/\alpha) \quad (1-1)$$

$$F(t; \alpha, \beta, \gamma) = 1 - \exp(-(t-\gamma)^\beta/\alpha) \quad (1-2)$$

The 3PW_{new} & 4PW Cdf's are:

$$F(t; \alpha, \beta, \gamma) = 1 - \exp(-(t^\beta - \gamma)/\alpha) \quad (1-3)$$

$$F(t; \alpha, \beta, \gamma_1, \gamma_2) = 1 - \exp(-((t-\gamma_1)^\beta - \gamma_2)/\alpha) \quad (1-4)$$

$$\alpha > 0; \beta > 0; \gamma, \gamma_1, \gamma_2 \geq 0.$$

The AB-model is:

$$y = a + b \cdot x + \epsilon \quad (1-5)$$

The two ABC-models are:

$$y = (a + b \cdot x)^c + \epsilon \quad (1-6)$$

$$y = a + b \cdot x^c + \epsilon \quad (1-7)$$

The AABC-model is:

$$y = a_1 + (a_2 + b \cdot x)^c + \epsilon \quad (1-8)$$

Notation

T	truncation-time of test
n, N	number of items [failed, on test], $n \leq N$
t_i	ordered individual failure-time i , $0 \leq t_i \leq T$, $i=1, 2, \dots, n$
F_i	$i/(N+1)$: s -unbiased estimate of individual empirical Cdf value
f_i	individual pdf value, $f_i \geq 0$
$F(\cdot), R(\cdot)$	[Cdf, Sf] of $(\cdot)$; $R(\cdot) = 1 - F(\cdot)$ for a continuous r.v.
F, F', K, K'	[F, F', K, K']-axis; $K = \ln(-\ln(1-F))$; $K = \ln(1-F')$; $K = \ln(K')$
α, β, γ	[scale, shape, location] parameter
a, b, c	arbitrary parameters

$\hat{}$ implies an estimate
 $\bar{}$ implies the average value.

Other, standard notation is given in "Information for Readers & Authors" at the rear of each issue.

2. TRANSFORMATIONS

The superiority of LRA/LLS is that we can do all kinds of transformations with a strong imagination, and can ignore whether the error ϵ in a regression model has a s -normal distribution. After a double logarithmic transformation, the 2PW Cdf (1-1) becomes a straight line (1-5), in which:

$$y = \ln(-\ln(R)), x = \ln(t), \quad (2-1)$$

$$a = -\ln(\alpha), b = \beta.$$

LRA/LLS can be applied to (2-1) directly, *eg*, Weibull (probability) paper is widely used in reliability practice. This magic really helps us get rid of nonlinear-system problems. The knack is that the calculation of the nonlinear system:

$$SS = \sum_{i=1}^n (F_i - (1 - \exp(-t_i^\beta/\alpha)))^2, \quad (2-2)$$

$$\partial SS/\partial \alpha = \partial SS/\partial \beta = 0.$$

is equivalent to the calculation of the linear system:

$$SS = \sum_{i=1}^n (y_i - (a + b \cdot x_i))^2, \quad (2-3)$$

$$\partial SS/\partial a = \partial SS/\partial b = 0.$$

SS = sum of squares of deviation from regression line.

The equivalence between (2-2) & (2-3) has a problem, widely studied in statistics concerning "super-accuracy", but the result is generally acknowledged and beyond the scope of this paper.

In view of the equivalence, the 3PW_{org} Cdf (1-2) can be transformed into 2 AB-models.

1. The conventional one (by a double logarithmic transformation):

$$y_1 = a_1 + b_1 \cdot x_1 \quad (\gamma \text{ is assumed known}):$$

$$y_1 = \ln(-\ln(R)), x_1 = \ln(t-\gamma), \quad (2-4)$$

$$a_1 = -\beta \cdot \ln(\alpha), b_1 = \beta.$$

2. A new one (by a single logarithmic transformation):

$$y_2 = a_2 + b_2 \cdot x_2 \quad (\beta \text{ is assumed known}):$$

$$y_2 = (-\ln(R))^{1/\beta}, x_2 = t, \quad (2-5)$$

$$a_2 = \gamma/\alpha, b_2 = 1/\alpha.$$

No evidence shows that (2-4) is better than (2-5). The main reason for preferring (2-4) is probably that (2-4) is similar to (2-1); (2-5) is ignored because of lack of imagination by others. Since the $3PW_{org}$ Cdf can become 2 (instead of 1) straight lines in 2-coordinate systems, LRA/LLS can be applied to it in special ways.

But the $3PW_{org}$ Cdf can become 1 (instead of 2) straight line,

$$y_3 = a_3 + b_3 \cdot x_3 \text{ (no parameter is assumed known):}$$

$$y_3 = R \cdot \ln(f/R), x_3 = t, \quad (2-6)$$

$$a_3 = -\gamma/\beta, b_3 = 1/\beta,$$

by a derivation transformation. The value of f_i can be estimated from t_i & F_i by, eg, any of (2-7),

$$f_{i,R} = (F_{i+1} - F_i)/(t_{i+1} - t_i), \quad (2-7a)$$

$$f_{i,L} = (F_i - F_{i-1})/(t_i - t_{i-1}), \text{ or} \quad (2-7b)$$

$$f_i \approx \frac{1}{2}(f_{i,R} + f_{i,L}), \quad (2-7c)$$

So LRA/LLS can be applied directly to the $3PW_{org}$ Cdf.

The $3PW_{org}$ Cdf can also become another straight line by an integral transformation. But it is not as simple as (2-6); so I prefer (2-6).

Since $\hat{\alpha}, \hat{\beta}$ can become non-positive (a knotty problem for other estimation methods), (2-4) & (2-5) have their flexible alternatives:

$$y_1 = \ln(-1/\ln(R)), x_1 = \ln(t-\gamma), \quad (2-8)$$

$$a_1 = -\beta \cdot \ln(-\alpha), b_1 = -\beta.$$

$$y_2 = (-1/\ln(R))^{-1/\beta}, x_2 = -t, \quad (2-9)$$

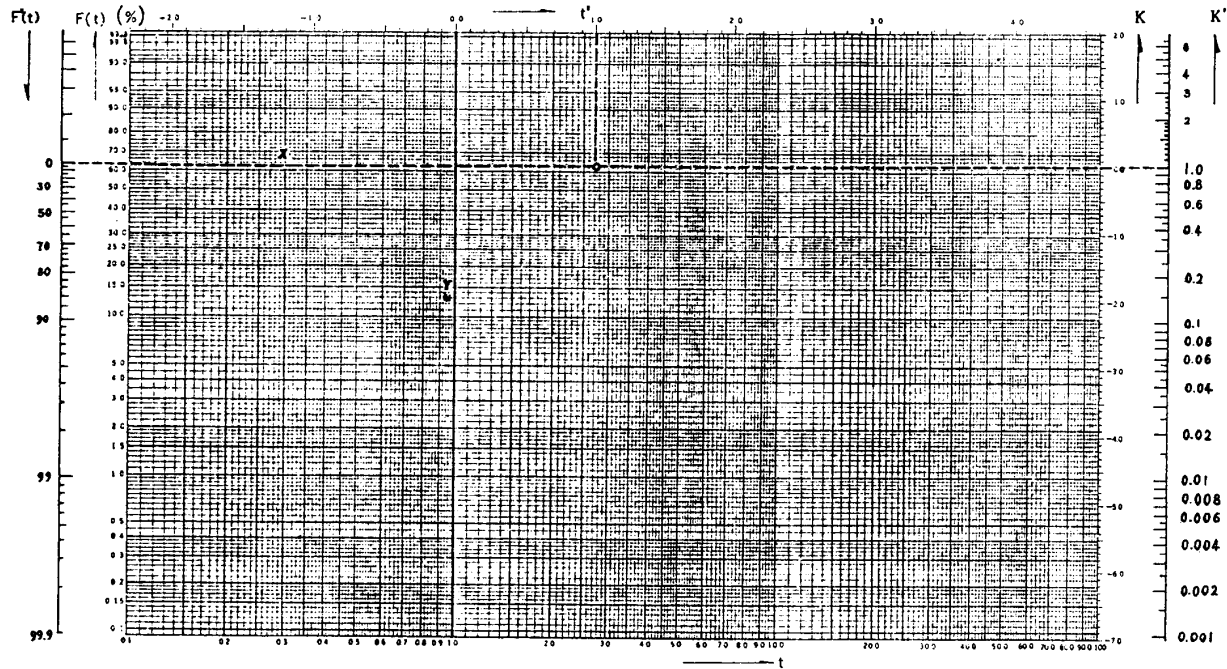
$$a_2 = -\gamma/\alpha, b_2 = -1/\alpha.$$

By using (2-4) - (2-5) and (2-8) - (2-9) in turn, the knotty problem can be observed closely and solved easily.

Similarly, imaginative transformations for the $3PW_{new}$ Cdf (1-3), the $4PW$ Cdf (1-4), and other regression models can also be found. It is always interesting to have a try at them.

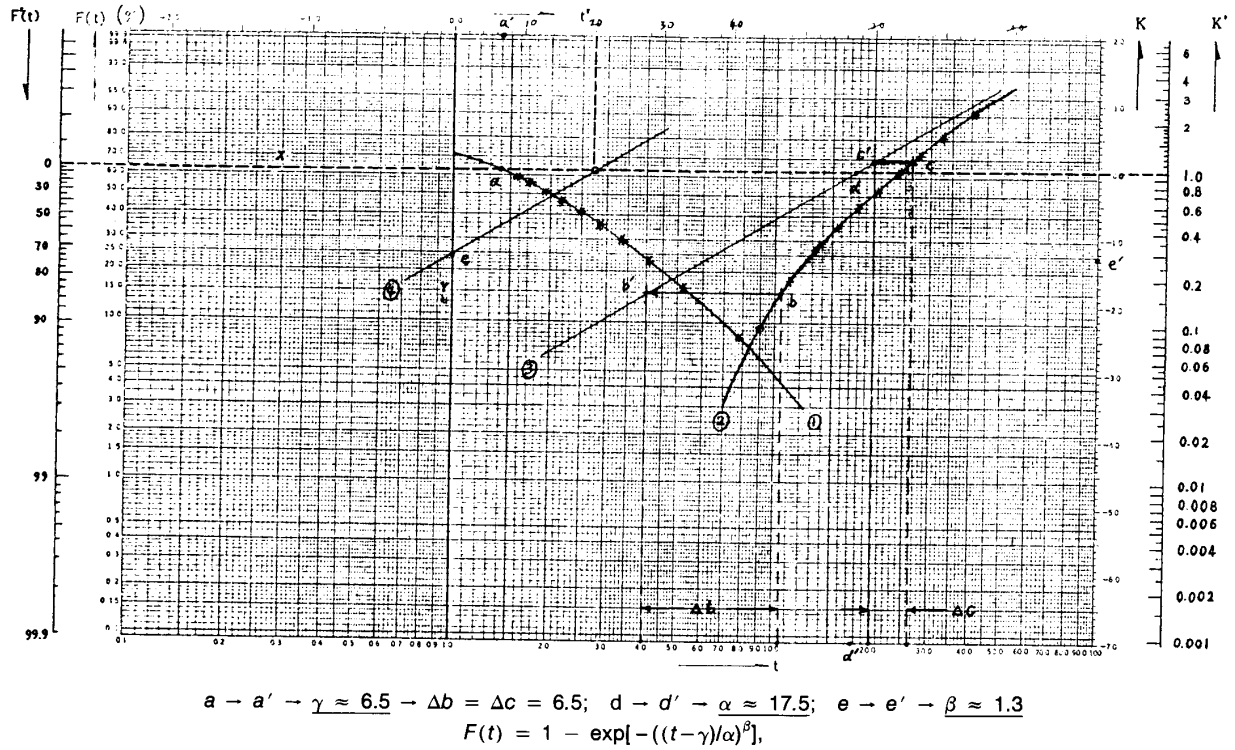
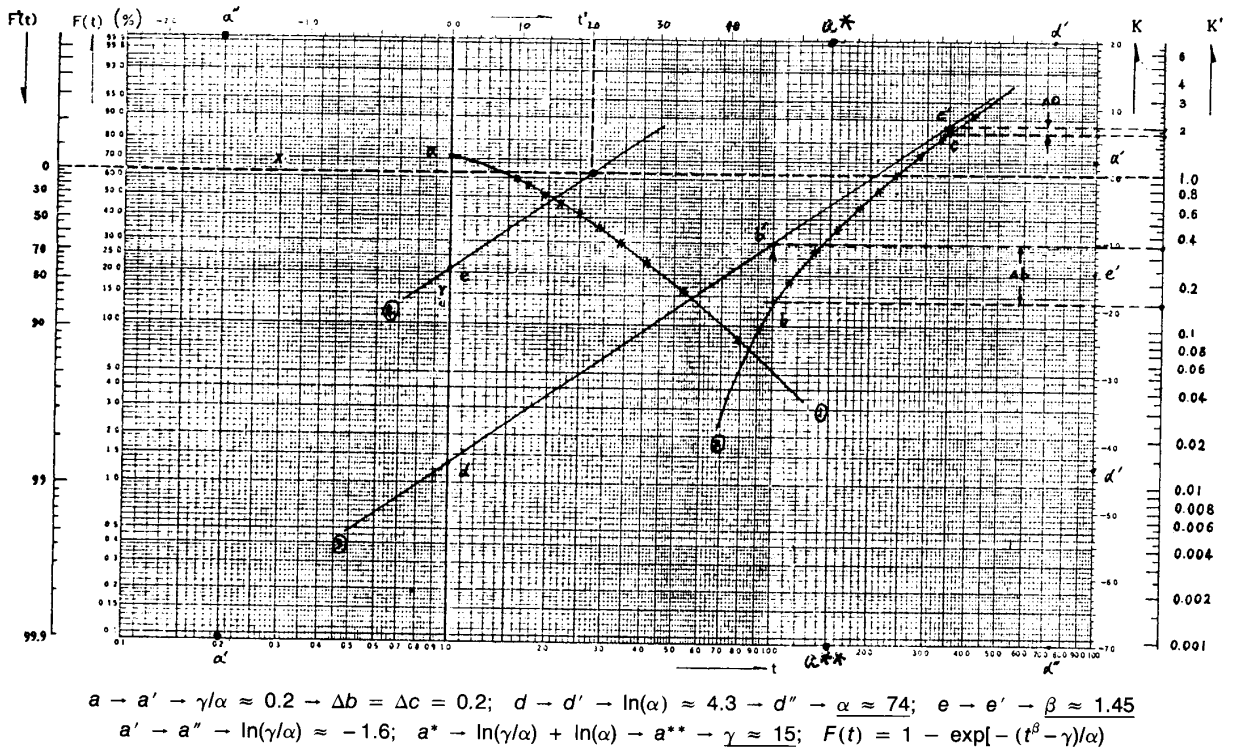
3. GRAPHICAL METHOD

Since the $3PW_{org}$ Cdf can become 2 straight lines in 2 different coordinate systems, the conventional graphical method and Weibull (probability) paper can be much improved. The basic improvement is to add another coordinate system, $y = -\ln(R)$, $x = t$, to the original Weibull paper (see figure 1a). The improved graphical method is briefly explained with an example (see figure 1b).



$$F(t) = 1 - \exp[-((t-\gamma)/\alpha)^\beta], F(t) = 1 - \exp[-(t^\beta - \gamma)/\alpha]$$

Figure 1a. Weibull Probability Paper: The Blank Paper

Figure 1b. Weibull Probability Paper: The 3PW_{org} PaperFigure 1c. Weibull Probability Paper: The 3PW_{new} Paper

Improved Graphical Method

1. Draw curve#1 & curve#2 with the 2 coordinate systems:

$$t \sim F(t) \text{ and } t' \sim F'(t)$$

2. Curve#1 intersects the X-axis at point a . Vertically, the point a' on the t' -axis shows $\gamma = 6.5$

3. On the t -axis, choose a vertical-line #1 at $t=4.0$; then another vertical-line #2 at $t = \gamma + 4.0 = 10.5$ is set. The latter intersects curve#2 at point b ; then point b' on the vertical-line #1 is set horizontally. Similarly, choose a vertical-line #3 at $t=20.0$, point c' is set on it. Connect b' & c' with straight-line#3; then the conventional graphical method can be used as shown in 4 & 5.

4. Straight-line#3 intersects the X-axis at point d . Vertically, the point d' on the t -axis shows $\alpha=17.5$.

5. Draw straight-line#4 passing through point o and parallel to straight-line#3; it intersects the Y-axis at point e . Horizontally, point e' on the K-axis shows $\beta=1.3$.

The improved paper can be used for the $3PW_{\text{new}}$ Cdf, ie, curve#1 & curve#2 can be used for the $3PW_{\text{new}}$ Cdf simultaneously. The graphical method for the $3PW_{\text{new}}$ Cdf is very similar to the improved graphical method for the $3PW_{\text{org}}$ Cdf (see figure 1c).

The improved Weibull paper and the improved graphical method can, of course, be improved further to make them even more convenient. But the principle is the same.

4. APPROXIMATE ANALYTIC METHOD

Notation

$$\phi_i = [\bar{x}_i \cdot \bar{x}_i - \bar{x}_i^2] / [\bar{x}_i \cdot \bar{y}_i - \bar{x}_i \cdot \bar{y}_i].$$

Since the $3PW_{\text{org}}$ Cdf can be a straight line for the approximation (2-7c), a simple formula can be derived from (2-4) & (2-6):

$$\hat{\gamma} = \bar{x}_3 - \bar{y}_3 \cdot \phi_3, \quad (4-1a)$$

$$\hat{\beta} = 1/\phi_1, \quad (4-1b)$$

$$\ln(\hat{\alpha}) = \bar{x}_1 - \bar{y}_1 \cdot \phi_1. \quad (4-1c)$$

From my experience, the errors⁴ in the 3 parameter-estimates caused by the approximation (2-9) are about 2% when $n \approx 10$. The error decreases as n increases. So 2% can be considered as the estimation error of (4-1).

⁴The error discussed here is the difference between the 'estimate calculated by this approximate method' and the 'estimate calculated by an 'exact' method described in section 2'. It does not refer to a comparison with the true value.

Another simple formula can be derived from (2-5) & (2-6) with the approximation (2-7c):

$$\hat{\alpha} = \phi_2, \quad (4-2a)$$

$$\hat{\beta} = 1/\phi_3, \quad (4-2b)$$

$$\hat{\gamma} = \bar{x}_2 - \bar{y}_2 \cdot \phi_2. \quad (4-2c)$$

Though (4-1) might have a little less error than (4-2), eq (4-2) is more convenient in dealing with the knotty problem ($\hat{\beta} < 0$) since $\hat{\beta}$ is estimated first. Relocating suspected outliers until the estimation result becomes satisfactory ($\hat{\beta} > 0$) can be a very useful practical reliability-engineering method, especially when handling field data, since the users can be more or less conscientious.

If less error is required, then I recommend the exact analytic method in section 5.

5. EXACT ANALYTIC METHOD

By applying LRA/LLS to the straight line (2-4) in which γ is assumed known, $\hat{\beta}$ is obtained, ie, (4-1b) or:

$$\hat{\beta} = g_1(\gamma). \quad (5-1)$$

Similarly, to (2-5) in which β is assumed known, $\hat{\gamma}$ is obtained, eg, (4-2c) or:

$$\hat{\gamma} = g_2(\beta). \quad (5-2)$$

In view of the equivalence, it is easy to understand that the intersection of the two curves (5-1) & (5-2) is the minimum solution. The iteration relationship can be analyzed; eg,

in (5-1) if $\gamma \rightarrow t_{\min}$, then $g_1(\gamma) \rightarrow -\infty$

in (5-2) if $\beta \rightarrow \infty$, then $g_2(\beta) \rightarrow -\infty$.

In addition, by applying LRA/LLS to the straight line (2-6), estimates $\hat{\beta}_R$ & $\hat{\beta}_L$ can be obtained from the approximations (2-7a) & (2-7b) respectively. So the initial iteration-limits are:

$$\beta_{\min} = \max(0, \min(\hat{\beta}_R, \hat{\beta}_L)) \quad (5-3a)$$

$$\beta_{\max} = \max(0, \hat{\beta}_R, \hat{\beta}_L) \quad (5-3b)$$

$$\text{Thus } \beta_{\min} \leq \beta \leq \beta_{\max}; \text{ (see figure 2)} \quad (5-4)$$

$$\Delta\beta \equiv \beta_{\max} - \beta_{\min}$$

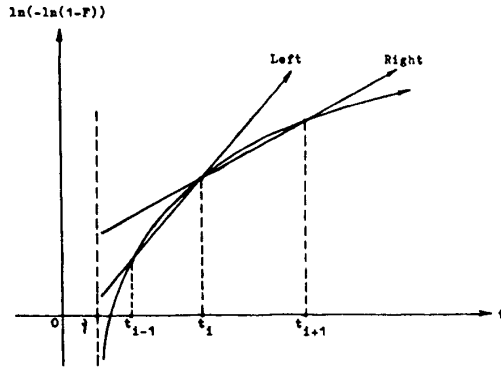


Figure 2. The "Right" & "Left" Approximations

My experience shows that if the curve of the $3PW_{org}$ Cdf is not too sharp, then $\Delta\beta < 2$ when $n \approx 10$. The $\Delta\beta$ decreases as n increases, so that usually $\Delta\beta \approx 1/2$. The narrow limits on β help to avoid the numerical iteration difficulties at multiple kinds of stationary points. I believe that, in practice, only 1 stationary point lies within $\Delta\beta$.

A simple algorithm is given here (see figure 3).

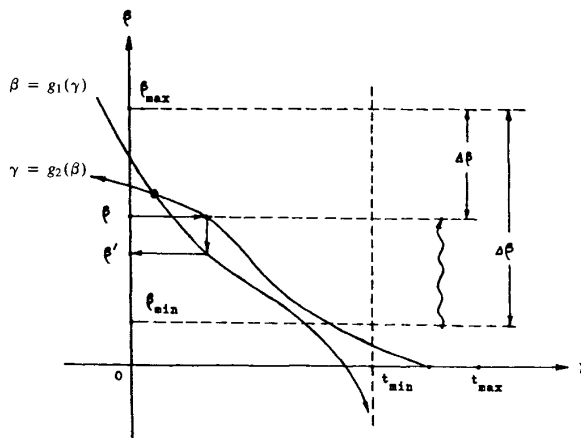


Figure 3. A Simple Idea for a Short Computer Program

Algorithm

1. Prepare the two iteration-limits (5-3).

Set $\delta = \text{"a small number such as 0.001"}$.

2. Set $\hat{\beta} = 1/2(\beta_{min} + \beta_{max})$; then calculate $\hat{\beta}' = g_1(g_2(\hat{\beta}))$.

3. If $\hat{\beta}' < \hat{\beta}$ then re-set $\beta_{min} = \hat{\beta}'$; else re-set $\beta_{max} = \hat{\beta}$. (So that $\Delta\beta$ is reduced rapidly.)

If $\Delta\beta > \delta$ then GoTo step 2.

4. Stop

Difficulties

D1. In practice $\Delta\beta$ might large (eg, $\Delta\beta > 5$) at the beginning because n is too small, or there is appreciable scatter in the data, or etc. No matter what the reason, $\hat{\beta}$ can still be calculated from the algorithm.

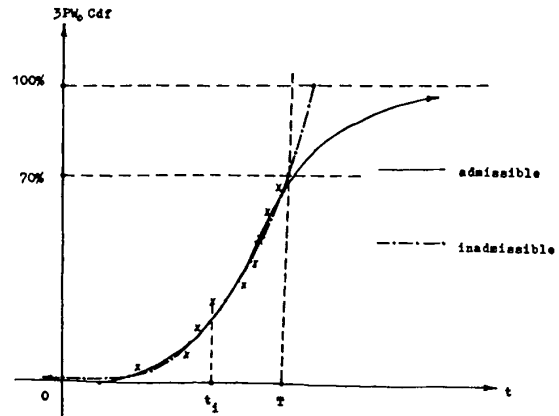


Figure 4. Admissible & Inadmissible Solutions

D2. The knotty one wherein $\beta_{min} = \beta_{max} = 0$. The principal reason for this situation is that the data are too censored to describe $3PW_{org}$ reasonably (see figure 4). Very unfortunately, this can rarely be predicted before an estimate has been made. This knotty problem can be quite bothersome for other estimation methods to deal with, since it is one of the inherent disadvantages of $3PW_{org}$ itself. But it can be solved easily here, since a local minimum (for the sum of squares) solution ($\hat{\beta} > 0$) can always be found and it is, of course, the optimization of the parameters. For example, the local minimum solution:

$$\hat{\beta} = 10.09, \hat{\alpha} = 17.89, \hat{\gamma} = 0.01$$

can replace the global minimum solution:

$$\hat{\beta} = -2, \hat{\alpha} = -2, \hat{\gamma} = 20$$

very well. To solve this knotty problem more reasonably, I recommend the following pragmatic procedure if there are enough data.

Pragmatic Procedure

1. Find an admissible local minimum solution to replace the inadmissible global minimum solution. This is usually satisfactory.

2. Use the admissible solution and its curve to spot the most-suspect potential outlier (often, the judgment about outliers is very subjective), then relocate it (for the moment).

3. Re-compute to see if the global minimum solution is admissible. If not, then GoTo step 2.

4. Stop.

In addition, I recommend an alternative distribution (see section 6) instead of an evasive estimation method. The alternative seems better in reliability practice (instead of in pure mathematics). Then why not try it in this case.

6. $3PW_{new}$ DISTRIBUTION

Two parameters are, sometimes, not enough in a Weibull distribution. Once a location parameter is added to the 2PW, the well-known $3PW_{org}$ is built. But another, unfamiliar one, the $3PW_{new}$, is also built. Though the latter seems to have been ignored, it has several obvious advantages.

• Reliability Meanings

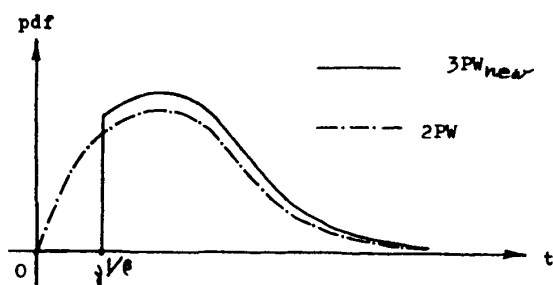


Figure 5a. The $3PW_{new}$ pdf vs the 2PW pdf

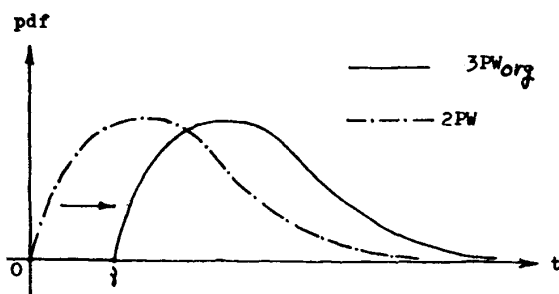


Figure 5b. The $3PW_{org}$ pdf vs the 2PW pdf

According to $3PW_{new}$, we know that, *eg*, the electronic components or parts which would fail during the time $0 < t < \hat{\gamma}^{1/\beta}$ in a life test have been screened out by efficient reliability controls (see figure 5a). Whereas according to $3PW_{org}$, it seems that, *eg*, a life test which should start at time $t=0$ (say, 8:00 am) as if started at another time $t=\hat{\gamma}$ (say, 8:00 pm) by a mistake (see figure 5b).

• Mathematical Characters

In $3PW_{new}$, $t > 0$ is always required even if $\hat{\gamma} < 0$ (see figure 6a). Whereas in the $3PW_{org}$ function, $t < 0$ is never restricted (see figure 6b); no matter what estimation methods can be developed, the problem $t < 0$ will never be solved.

• Smoothing Similarity

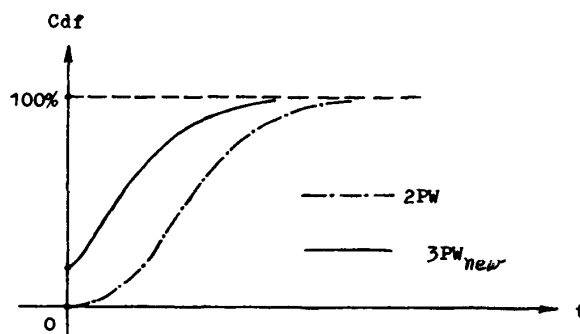


Figure 6a. The $3PW_{new}$ Cdf vs the 2PW Cdf

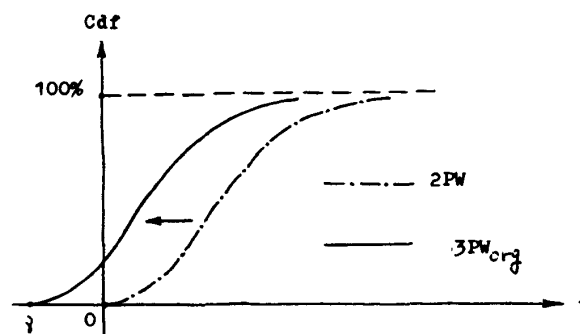


Figure 6b. The $3PW_{org}$ Cdf vs the 2PW Cdf

The $3PW_{new}$ Cdf is quite similar to the $3PW_{org}$ Cdf (see figure 7), and both are exactly the same when $\hat{\gamma}=0$ or $\hat{\beta}=1$. The difference between the two Cdf's is usually much smaller than the errors in practical data (especially, in field data). So $3PW_{new}$ can be used any place where $3PW_{org}$ is well used, and even replace the $3PW_{org}$ when it "fails".

• Estimation Convenience

Because the estimation methods for the $3PW_{new}$ are extremely similar to those for the $3PW_{org}$ (the graphic method is an example given above), and because the $3PW_{new}$ pdf is exactly the same as the 2PW pdf except for a proportional-constant, much estimation convenience can be anticipated.

• Distribution Advancement

The $3PW_{new}$ can be combined with $3PW_{org}$ to form a robust 4PW with 2 location parameters. The old "unsolvable" problem, $\hat{\gamma} < 0$, in $3PW_{org}$ can now be solved by replacing $3PW_{org}$ with 4PW in which, $\hat{\gamma}_2 = (-\hat{\gamma}_1)^\beta$ since those

electronic components or parts which would fail during $\hat{\gamma} < t < 0$ or $-\infty < t < 0$ have certainly been screened-out before a life test (see figure 8).

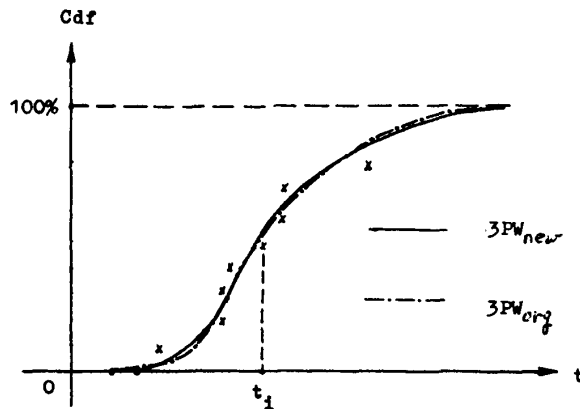


Figure 7. The $3PW_{new}$ Cdf vs the $3PW_{org}$ Cdf

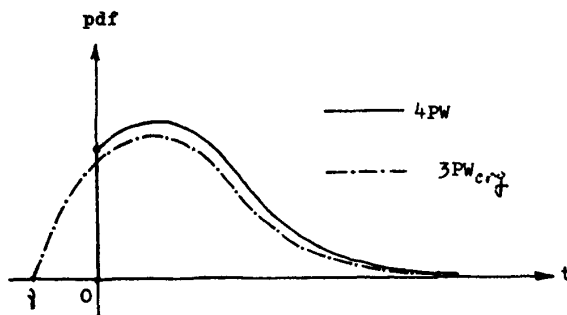


Figure 8. The $4PW$ pdf vs the $3PW_{org}$ pdf

7. ABC-MODELS

There are many kinds of 2-parameter nonlinear regression models such as $y = a \cdot x + b$, or $y = a + \ln(1 + b \cdot x)$. All of them can be transformed into the AB-model to which LRA/LLS can be applied directly. Similarly, all kinds of 3-parameter nonlinear regression models can be transformed into the 2 ABC-models, to which GLRA/GLLS can be applied identically. The $3PW_{org}$ Cdf is still used as the example.

Algorithm

1. Transform the $3PW_{org}$ Cdf (1-2) into the ABC-model (1-6):

$$y = -\ln(1-F), x=t; \quad (7-1a)$$

$$a = -\gamma/\alpha, b=1/\alpha, c = \beta. \quad (7-1b)$$

2a. Transform the ABC-model into 2 AB-models:

$$y_1 = \ln(y), x_1 = \ln(x+a/b), \quad (7-2a)$$

$$a_1 = c \cdot \ln(b), b_1 = c; \quad (7-2b)$$

$$y_2 = y^{1/c}, x_2 = x, \quad (7-3a)$$

$$a_2 = a, b_2 = b; \quad (7-3b)$$

2b. Transform the ABC-model into a real AB-model:

$$y_3 = y/y', x_3 = x, \quad (7-4a)$$

$$a_3 = a/(b \cdot c), b_3 = 1/c. \quad (7-4b)$$

3. Obtain the data y'_i from the data x_i & y_i by:

$$y'_{i,R} \equiv (y_{i+1} - y_i)/(x_{i+1} - x_i), \quad (7-5a)$$

$$y'_{i,L} \equiv (y_i - y_{i-1})/(x_i - x_{i-1}), \text{ or} \quad (7-5b)$$

$$y'_i \approx \frac{1}{2}(y'_{i,R} + y'_{i,L}). \quad (7-5c)$$

4. Obtain $\hat{c}_R$ with (7-5a), and $\hat{c}_L$ with (7-5b). Set up the initial iteration-limits:

$$c_{\min} = \max(0, \min(\hat{c}_R, \hat{c}_L)), \quad (7-6a)$$

$$c_{\max} = \max(0, \hat{c}_R, \hat{c}_L); \quad (7-6b)$$

$$\text{thus } \hat{c}_{\min} < \hat{c} < \hat{c}_{\max}. \quad (7-7)$$

If $c_{\min}=c_{\max}=0$ then put $y = 1/y, x = -x$.

5. Estimate $\hat{a}, \hat{b}, \hat{c}$ by one of the 3 methods; then —

$$\hat{\alpha} = 1/\hat{b}, \hat{\beta}=\hat{c}, \hat{\gamma}=-\hat{a}/\hat{b}.$$

This algorithm summarizes the contents of this paper. ◀

Similarly, the $3PW_{new}$ Cdf (1-3) can be transformed into the other ABC-model (1-7).

The 2 ABC-models are 2 elementary 3-parameter nonlinear regression models. They can be used to form large multiparameter nonlinear regression models. For example, the AABC-model (1-8) is directly built on the 2 ABC-models, *ie*, it will be (1-6) when $a_1=0$; and (1-7) when $a_2=0$. So it can be analyzed based upon the analyses of the 2 ABC-models, though a few other techniques are still necessary.

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(τ, k) Block Replacement Policy with Idle Count

The following changes/corrections are needed in [1].

p 563, col 1, line 1 ↓

Lemma 1. For $r = 1, \dots, N$,

p 563, col 2, line 8 ↓

Theorem 2. Let $\bar{C}^*(\cdot)$ be the optimal cost rate of BRO(T)

p 563, col 2, line 9 ↑

$$C_N^*(\cdot, \cdot) < \bar{C}^*(\cdot) \leq \bar{C}(\infty) = c/\mu = C_N(\infty, \cdot)$$

p 564, col 1, lines 18-16 ↓

ing sequence of k : $0 \leq d^* < \theta_N(1)$; $\theta_N(k) < d^* < \theta_N(k+1)$

for some k , $1 \leq k < N$; $d^* = \theta_N(k)$ for some k , $1 \leq k \leq$

N ; and $d^* > \theta_N(N)$

p 564, col 2, line 15 ↓

= 2 with the corresponding cost rate,

$$C_N^*(\cdot, \cdot) = \$2.73/\text{year}.$$

p 565, line 2, title of Table 1

Cost Performance of BRIC(τ, k), BRIP(T, d), and BRO(T) with Effect of the Fleet Size and Downtime Cost

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- [1] K.S. Park, Y.K. Yoo, " (τ, k) Block replacement policy with idle count", *IEEE Trans. Reliability*, vol 42, 1993 Dec, pp 561-565. (Original IEEE Log Number 92-07398)

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